

$\lambda\phi^4$ theory and Feynman rules in position space

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Abstract

The position-space Feynman rules are derived for the two-point correlator of real scalar $\lambda\phi^4$ theory. The Schrödinger, Heisenberg, and interaction pictures are first defined, distinguishing free evolution from full evolution. Nested commutators and the Baker–Campbell–Hausdorff formula are used to reconstruct the free-field mode expansion and obtain the interaction Hamiltonian.

The Gell-Mann–Low formula, the Dyson series, and Wick’s theorem translate contractions into propagators, vertices, and spacetime integrals. The cancellation of vacuum bubbles is explained, first-order contributions are calculated, and the three connected second-order topologies are presented with their integrals and symmetry factors. Figures and examples connect field evolution to operator calculations and show how diagrammatic rules emerge from the combinatorics of contractions. Loop integrals remain formal; their regularization and renormalization are not developed.

1 Introduction

The passage from quantum-field dynamics to Feynman rules requires bringing together several structures: the evolution of states and operators, the choice of vacuum, time ordering, and the combinatorics of contractions. The simplicity of a diagram is the result of this work. Understanding how it is constructed reveals what each line represents, which operation a vertex introduces, and where the coefficient accompanying a contribution comes from. These notes develop that construction for the two-point correlator of real scalar $\lambda\phi^4$ theory in position space.

The first task is to identify the operators used in the calculation. The Schrödinger, Heisenberg, and interaction pictures share initial data but distribute time evolution differently between states and operators. Distinguishing the free Hamiltonian from the full Hamiltonian establishes the role of ϕ_I and its relation to the interacting field ϕ_H . Nested commutators and the Baker–Campbell–Hausdorff formula reconstruct the mode expansion, explicitly verifying the free evolution underlying the perturbative development and yielding the interaction Hamiltonian.

The second task is to turn this operator structure into an organized calculation. The Gell-Mann–Low formula relates the interacting-vacuum correlator to expectation values in the free vacuum; the Dyson series organizes interaction insertions, and Wick’s theorem reduces them to products of propagators. We follow the contractions through the first-order contributions and the three connected second-order topologies, including their integrals and symmetry factors. Normalization establishes which pieces cancel: vacuum bubbles disappear, while loops attached to the external points remain.

The central result is a justification of the Feynman rules from the canonical formalism and contraction counting. Once this foundation is established, subsequent calculations can be organized through diagrams without reconstructing the full operator expansion in each case. We use $\hbar = c = 1$, the metric $(+, -, -, -)$, and an interaction without normal ordering, working around the symmetric vacuum. Loop integrals remain formal; their evaluation, regularization, and renormalization constitute the next stage of the analysis.

2 $\lambda\phi^4$ theory

We consider a real scalar field in the perturbative regime around the symmetric vacuum, with $m^2 > 0$ and $\lambda \geq 0$. Its Lagrangian is

$$\mathcal{L} := \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{m^2}{2} \phi^2 - \frac{\lambda}{4!} \phi^4. \quad (1)$$

The dot over the field denotes its partial derivative with respect to time, with the spatial coordinate held fixed:

$$\dot{\phi}(t, \mathbf{x}) := \partial_t \phi(t, \mathbf{x}) = \frac{\partial \phi(t, \mathbf{x})}{\partial t} = \partial_0 \phi(t, \mathbf{x}). \quad (2)$$

With the metric convention $(+, -, -, -)$, the canonical momentum conjugate to the field is

$$\pi(t, \mathbf{x}) := \frac{\partial \mathcal{L}}{\partial \dot{\phi}(t, \mathbf{x})} = \dot{\phi}(t, \mathbf{x}). \quad (3)$$

The Hamiltonian density is

$$\mathcal{H} := \frac{1}{2} \pi^2 + \frac{1}{2} |\nabla \phi|^2 + \frac{m^2}{2} \phi^2 + \frac{\lambda}{4!} \phi^4. \quad (4)$$

Integrating the Hamiltonian density over space, we separate the free part from the interaction:

$$\begin{aligned} H(t) &:= \int d^3x \mathcal{H}(t, \mathbf{x}) \\ &= \underbrace{\int d^3x \left[\frac{1}{2} \pi^2(t, \mathbf{x}) + \frac{1}{2} |\nabla \phi(t, \mathbf{x})|^2 + \frac{m^2}{2} \phi^2(t, \mathbf{x}) \right]}_{H_0(t) \text{ free part}} + \underbrace{\frac{\lambda}{4!} \int d^3x \phi^4(t, \mathbf{x})}_{H'(t) \text{ interaction}}. \end{aligned} \quad (5)$$

The time arguments above indicate that the dynamical variables are evaluated at a given instant; they do not represent explicit time dependence of the Hamiltonian. When moving to the operator formalism, we will fix the splitting $H = H_0 + H'_S$ using the initial operators at $t = 0$.

3 Dynamical pictures

We use $\hbar = c = 1$ and take $t_0 = 0$. We consider a theory whose Hamiltonian does not depend explicitly on time.

3.1 Fields and evolution generators

Definition 1 (Field and conjugate momentum in the Schrödinger picture). We fix the initial operators of the field and its conjugate momentum:

$$\boxed{\phi_S(\mathbf{x}) := \phi(0, \mathbf{x}), \quad \pi_S(\mathbf{x}) := \pi(0, \mathbf{x}).} \quad (6)$$

The subscript S denotes the Schrödinger picture. These operators carry no time argument; they represent the initial data of the full theory. These constructions are used in the perturbative canonical formalism; products of fields at the same point require regularization.

Definition 2 (Hamiltonians in the Schrödinger picture). Using these operators, we define the free part and the interaction term:

$$\boxed{H_0 := \int d^3x \left[\frac{1}{2} \pi_S^2(\mathbf{x}) + \frac{1}{2} |\nabla \phi_S(\mathbf{x})|^2 + \frac{m^2}{2} \phi_S^2(\mathbf{x}) \right].} \quad (7)$$

$$\boxed{H'_S := \frac{\lambda}{4!} \int d^3x \phi_S^4(\mathbf{x}), \quad H = H_0 + H'_S.} \quad (8)$$

Thus, H , H_0 , and H'_S are fixed Schrödinger operators.

Definition 3 (Time-evolution operators). For the time-independent Hamiltonians H and H_0 ¹, we define the full and free time-evolution operators, respectively, by

$$U(t) := e^{-iHt}, \quad U_0(t) := e^{-iH_0t}. \quad (9)$$

Definition 4 (Field in the Heisenberg picture). Starting from the initial operator $\phi_S(\mathbf{x})$, we define the interacting field in the Heisenberg picture using the full Hamiltonian $H = H_0 + H'_S$:

$$\boxed{\phi_H(t, \mathbf{x}) := e^{iHt} \phi_S(\mathbf{x}) e^{-iHt}.} \quad (10)$$

In $\lambda\phi^4$ theory, this is the interacting field and describes the evolution of the full theory.

¹For a fixed Hamiltonian, the Schrödinger equation has the solution $|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle$. The condition $\partial H / \partial t = 0$ means that the Hamiltonian has no explicit time dependence. This idea already appears in familiar examples from classical mechanics:

$$\begin{aligned} H_{\text{free}}(\mathbf{x}, \mathbf{p}) &= \frac{\mathbf{p}^2}{2m} && \text{(free particle),} \\ H_{\text{osc}}(x, p) &= \frac{p^2}{2m} + \frac{1}{2} k x^2 && \text{(harmonic oscillator),} \\ H_{\text{Coul}}(\mathbf{x}, \mathbf{p}) &= \frac{\mathbf{p}^2}{2m} - \frac{\kappa}{r} && \text{(Coulomb potential),} \\ H_{\text{grav}}(z, p_z) &= \frac{p_z^2}{2m} + mgz && \text{(uniform gravity),} \\ H_{\text{pend}}(\theta, p_\theta) &= \frac{p_\theta^2}{2m\ell^2} + mg\ell(1 - \cos \theta) && \text{(simple pendulum).} \end{aligned} \quad (I)$$

Here m is the mass, $k > 0$ the spring constant, $\kappa > 0$ the strength of the attraction, and $r = |\mathbf{x}|$. In uniform gravity, g is the gravitational acceleration, z the upward height coordinate, and p_z its conjugate momentum. For the pendulum, ℓ is the fixed length, θ the angle from the downward vertical, and $p_\theta = m\ell^2\dot{\theta}$ its conjugate momentum; the suspension point is fixed. We assume no friction. With these parameters constant, none of the five Hamiltonians contains t explicitly, even though position and momentum change during motion. More generally, $H = \mathbf{p}^2/(2m) + V(\mathbf{x})$ satisfies this condition for any static potential. A time-independent Hamiltonian can therefore describe both motion and interaction.

Definition 5 (Field in the interaction picture). Starting from the same initial operator $\phi_S(\mathbf{x})$, we define the field in the interaction picture using the free part H_0 :

$$\boxed{\phi_I(t, \mathbf{x}) := e^{iH_0 t} \phi_S(\mathbf{x}) e^{-iH_0 t}.} \quad (11)$$

The subscript I denotes the interaction picture; this field evolves freely.²

Remark 1. The equality $\phi_S(\mathbf{x}) = \phi_H(0, \mathbf{x})$ does not imply that the interacting field is free: $\phi_H(0, \mathbf{x})$ is the full field of $\lambda\phi^4$ theory evaluated at the initial time. At $t = 0$, the Schrödinger, Heisenberg, and interaction pictures share the same field operator,

$$\phi_S(\mathbf{x}) = \phi_H(0, \mathbf{x}) = \phi_I(0, \mathbf{x}). \quad (12)$$

If these initial data are evolved with H_0 , we obtain the free field $\phi_I(t, \mathbf{x})$. If they are evolved with the full Hamiltonian $H = H_0 + H'_S$, we obtain the full field $\phi_H(t, \mathbf{x})$. Thus, in general, $\phi_H(t, \mathbf{x}) \neq \phi_I(t, \mathbf{x})$ if $t \neq 0$.

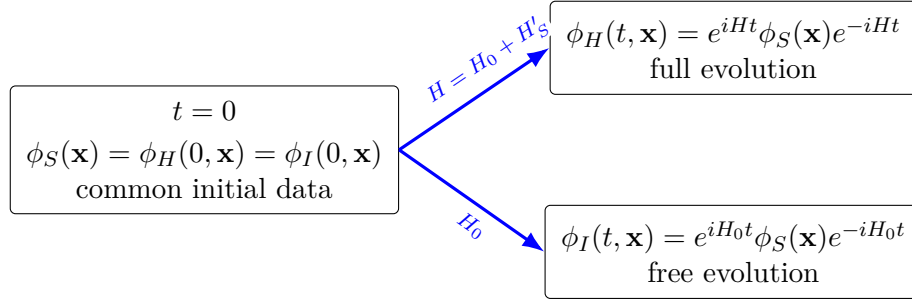


Figure 1. Summary of (10) and (11): the same initial data evolve with H to give ϕ_H and with H_0 to give ϕ_I .

3.2 States and operators in each picture

Definition 6 (Schrödinger picture). States evolve with the full Hamiltonian, while operators without explicit time dependence remain fixed:

$$\boxed{|\psi_S(t)\rangle := e^{-iHt} |\psi_S(0)\rangle, \quad \frac{\partial A_S}{\partial t} = 0.} \quad (13)$$

Therefore, the operator is denoted by A_S , without a time argument; it is not written as $A_S(t)$.

Definition 7 (Heisenberg picture). States are constant and operators evolve with the full Hamiltonian:

$$\boxed{|\psi_H\rangle := |\psi_S(0)\rangle, \quad A_H(t) := e^{iHt} A_S e^{-iHt}.} \quad (14)$$

Definition 8 (Interaction picture). Operators evolve with H_0 :

$$\boxed{A_I(t) := e^{iH_0 t} A_S e^{-iH_0 t}.} \quad (15)$$

From the definition of the Heisenberg picture,

$$A_H(t) = e^{iHt} A_S e^{-iHt}, \quad (16)$$

²We follow the terminology of D. Tong, *Lectures on Quantum Field Theory*, Sections 3.1 and 3.7, which distinguish the Heisenberg-picture field of the full theory from the interaction-picture field. See [Interacting Fields](#).

we solve for

$$A_S = e^{-iHt} A_H(t) e^{iHt}. \quad (17)$$

Substituting this result into (15),

$$A_I(t) = e^{iH_0 t} e^{-iHt} A_H(t) e^{iHt} e^{-iH_0 t}. \quad (18)$$

$$A_I(t) = U_I(t) A_H(t) U_I^\dagger(t). \quad (19)$$

where

$$U_I(t) := e^{iH_0 t} e^{-iHt}. \quad (20)$$

Applied to the field, relation (18) shows that the transformation to the interaction picture cancels the full evolution, leaving only free evolution:

$$\begin{aligned} \phi_I(t, \mathbf{x}) &= e^{iH_0 t} e^{-iHt} \phi_H(t, \mathbf{x}) e^{iHt} e^{-iH_0 t} \\ &= e^{iH_0 t} e^{-iHt} \left(e^{iHt} \phi_S(\mathbf{x}) e^{-iHt} \right) e^{iHt} e^{-iH_0 t} \\ &= e^{iH_0 t} \phi_S(\mathbf{x}) e^{-iH_0 t}. \end{aligned} \quad (21)$$

Equivalently,

$$A_H(t) = U_I^\dagger(t) A_I(t) U_I(t). \quad (22)$$

States carry the remaining evolution:

$$|\psi_I(t)\rangle := e^{iH_0 t} |\psi_S(t)\rangle, \quad i \frac{d}{dt} |\psi_I(t)\rangle = H'_I(t) |\psi_I(t)\rangle, \quad (23)$$

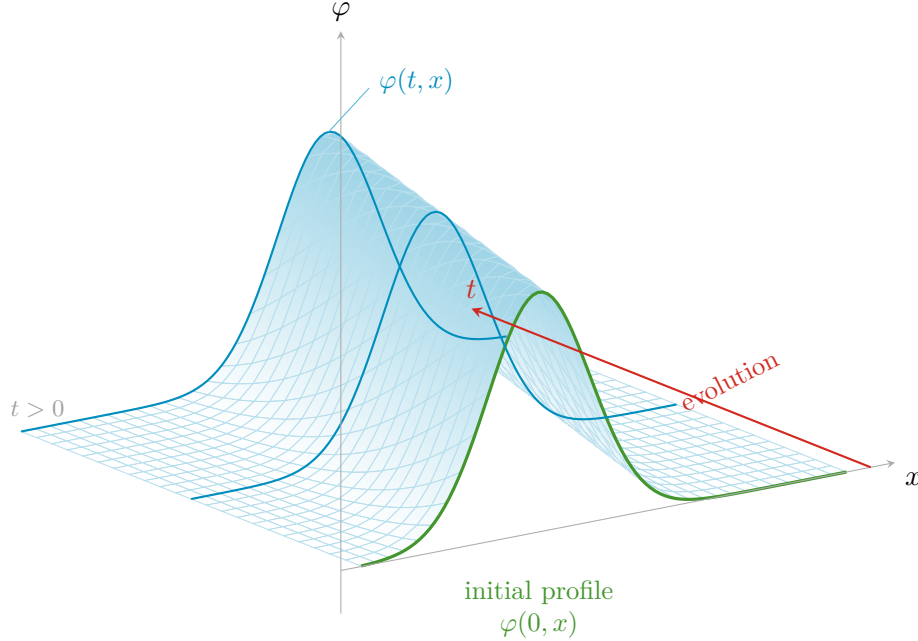


Figure 2. From the spatial profile to its time evolution. The green curve represents an initial profile at $t = 0$; the blue slices show later profiles. The red axis indicates time, and height indicates the value of the field. The surface is a schematic classical profile, not a graph of the quantum operator or a computed solution of $\lambda\phi^4$ theory.

The figure illustrates how a spatial profile extends through time. In the operator definitions, the same time appears in both ϕ_I and ϕ_H ; their evolutions differ through the generator, H_0 or H .

A concrete example in one spatial dimension is the free mode

$$\varphi(t, x) = A \cos(kx - \omega_k t), \quad \omega_k = \sqrt{k^2 + m^2}.$$

At $t = 0$ we have the profile $A \cos(kx)$; at later times, the phase $-\omega_k t$ shifts the crests and troughs. This wave is an analytical example of free evolution; the surface in the drawing merely illustrates the geometric construction of a profile over time.

4 Interaction Hamiltonian

To express $H'_I(t)$ in terms of free fields, we first verify, using the mode expansion, the evolution with H_0 defined in (11).

4.1 Reconstructing the free field

Example 1 (Verification of free evolution). We write the common initial operator $\phi_S(\mathbf{x}) = \phi_I(0, \mathbf{x})$ in the mode expansion associated with the free Hamiltonian H_0 :

$$\phi_S(\mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left[a_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (24)$$

Here $\omega_{\mathbf{k}} := \sqrt{\mathbf{k}^2 + m^2}$. The operators $a_{\mathbf{k}}$ and $a_{\mathbf{k}}^\dagger$ are defined with respect to this free part, whose vacuum satisfies $a_{\mathbf{k}} |0\rangle = 0$. Consider the free Hamiltonian, omitting the zero-point energy constant,

$$H_0 = \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}}, \quad (25)$$

and the commutation relations

$$[a_{\mathbf{k}}, a_{\mathbf{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}), \quad [a_{\mathbf{k}}, a_{\mathbf{p}}] = [a_{\mathbf{k}}^\dagger, a_{\mathbf{p}}^\dagger] = 0. \quad (26)$$

Show, using nested commutators and the Baker–Campbell–Hausdorff formula, that

$$e^{iH_0 t} \phi_S(\mathbf{x}) e^{-iH_0 t} = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left[a_{\mathbf{k}} e^{-i\omega_{\mathbf{k}} t + i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}} t - i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (27)$$

Conclude that the evolution generated by H_0 reconstructs the free Klein–Gordon field $\phi_{\text{libre}}(t, \mathbf{x})$.

Solution. According to the definition of the interaction picture, we compute

$$\phi_I(t, \mathbf{x}) = e^{iH_0 t} \phi_S(\mathbf{x}) e^{-iH_0 t}. \quad (28)$$

We first obtain the elementary commutators³

$$[H_0, a_{\mathbf{k}}] = -\omega_{\mathbf{k}} a_{\mathbf{k}}, \quad [H_0, a_{\mathbf{k}}^\dagger] = \omega_{\mathbf{k}} a_{\mathbf{k}}^\dagger. \quad (29)$$

³We use $[AB, C] = A[B, C] + [A, C]B$. Entonces

$$\begin{aligned} [H_0, a_{\mathbf{k}}] &= \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} [a_{\mathbf{p}}^\dagger a_{\mathbf{p}}, a_{\mathbf{k}}] \\ &= \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} (a_{\mathbf{p}}^\dagger [a_{\mathbf{p}}, a_{\mathbf{k}}] + [a_{\mathbf{p}}^\dagger, a_{\mathbf{k}}] a_{\mathbf{p}}) \\ &= - \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{k}) a_{\mathbf{p}} = -\omega_{\mathbf{k}} a_{\mathbf{k}}. \end{aligned} \quad (\text{II})$$

We now apply the Baker–Campbell–Hausdorff conjugation formula (also called Hadamard’s lemma) with $A = iH_0t$ and $B = \phi_S(\mathbf{x}) = \phi_I(0, \mathbf{x})$:

$$\boxed{e^{iH_0t} B e^{-iH_0t} = B + it[H_0, B] + \frac{(it)^2}{2!}[H_0, [H_0, B]] + \frac{(it)^3}{3!}[H_0, [H_0, [H_0, B]]] + \dots} \quad (30)$$

Let us now compute the commutators appearing in (30). Using (29) in expansion (24), the first commutator is

$$[H_0, \phi_S(\mathbf{x})] = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left[-\omega_{\mathbf{k}} a_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} + \omega_{\mathbf{k}} a_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (31)$$

Commuting once more with H_0 ,

$$[H_0, [H_0, \phi_S(\mathbf{x})]] = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left[\omega_{\mathbf{k}}^2 a_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} + \omega_{\mathbf{k}}^2 a_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (32)$$

The third commutator already reveals the alternating signs:

$$[H_0, [H_0, [H_0, \phi_S(\mathbf{x})]]] = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left[-\omega_{\mathbf{k}}^3 a_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} + \omega_{\mathbf{k}}^3 a_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (33)$$

Thus, for n nested commutators,

$$\underbrace{[H_0, [H_0, \dots [H_0, \phi_S(\mathbf{x})] \dots]]}_{n \text{ times}} = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left[(-\omega_{\mathbf{k}})^n a_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} + \omega_{\mathbf{k}}^n a_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \right]. \quad (34)$$

Substituting these commutators into (30),

$$\begin{aligned} \phi_I(t, \mathbf{x}) &:= e^{iH_0t} \phi_S(\mathbf{x}) e^{-iH_0t} \\ &= \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left[a_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} \sum_{n=0}^{\infty} \frac{(-i\omega_{\mathbf{k}}t)^n}{n!} + a_{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \sum_{n=0}^{\infty} \frac{(i\omega_{\mathbf{k}}t)^n}{n!} \right] \\ &= \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left[a_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} \right] \equiv \phi_{\text{libre}}(t, \mathbf{x}). \end{aligned} \quad (35)$$

The last equality defines the notation $\phi_{\text{libre}}(t, \mathbf{x})$ for the free Klein–Gordon field given by this mode expansion. Therefore,

$$\boxed{\phi_I(t, \mathbf{x}) = \phi_{\text{libre}}(t, \mathbf{x})}. \quad (36)$$

Summing the commutators reconstructs the phases $e^{\mp i\omega_{\mathbf{k}}t}$ and verifies the mode expansion of the free field stated in the problem. □

Remark 2. The field ϕ_I is not a “part” of ϕ_H in an additive sense. It is the free field with which we organize the effect of $H'_I(t)$ perturbatively.

and similarly

$$\begin{aligned} [H_0, a_{\mathbf{k}}^\dagger] &= \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} [a_{\mathbf{p}}^\dagger a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] \\ &= \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} a_{\mathbf{p}}^\dagger [a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] = \omega_{\mathbf{k}} a_{\mathbf{k}}^\dagger. \end{aligned} \quad (\text{III})$$

4.2 Constructing the interaction

Having computed the field ϕ_I , we transform a different operator: the interaction term H'_S defined in (8). Recall that

$$H'_S = \frac{\lambda}{4!} \int d^3 z \phi_S^4(\mathbf{z}). \quad (37)$$

Its expression in the interaction picture is

$$\boxed{H'_I(t) := e^{iH_0 t} H'_S e^{-iH_0 t}.} \quad (38)$$

We now use the result of Example 1, (36): $e^{iH_0 t} \phi_S(\mathbf{z}) e^{-iH_0 t} = \phi_I(t, \mathbf{z})$. Conjugation preserves products: we insert $e^{-iH_0 t} e^{iH_0 t} = 1$ between the four factors of ϕ_S^4 and apply this result to each factor. Substituting (8),

$$\begin{aligned} e^{iH_0 t} H'_S e^{-iH_0 t} &= \frac{\lambda}{4!} \int d^3 z e^{iH_0 t} \phi_S^4(\mathbf{z}) e^{-iH_0 t} \\ &= \frac{\lambda}{4!} \int d^3 z \left(e^{iH_0 t} \phi_S(\mathbf{z}) e^{-iH_0 t} \right)^4 \\ &= \frac{\lambda}{4!} \int d^3 z \phi_I^4(t, \mathbf{z}). \end{aligned} \quad (39)$$

Therefore,

$$\boxed{H'_I(t) = e^{iH_0 t} H'_S e^{-iH_0 t} = \frac{\lambda}{4!} \int d^3 z \phi_I^4(t, \mathbf{z}).} \quad (40)$$

Only the interaction term appears here because we start from H'_S . When transforming the total Hamiltonian, the free part is also present:

$$e^{iH_0 t} H e^{-iH_0 t} = H_0 + H'_I(t). \quad (41)$$

For states, however, the change of picture absorbs the free evolution. Differentiating $|\psi_I(t)\rangle = e^{iH_0 t} |\psi_S(t)\rangle$ gives

$$\begin{aligned} i \frac{d}{dt} |\psi_I(t)\rangle &= \left[-H_0 + e^{iH_0 t} H e^{-iH_0 t} \right] |\psi_I(t)\rangle \\ &= H'_I(t) |\psi_I(t)\rangle. \end{aligned} \quad (42)$$

The term $-H_0$ comes from differentiating $e^{iH_0 t}$ and cancels the free part of (41). This is why the Dyson series, which describes the evolution of these states, uses $H'_I(t)$.

The interaction is used without normal ordering: the term ϕ_I^4 is not replaced by $:\phi_I^4:$, so the local contractions that give rise to the loops in the examples are retained.

The integral in (40) runs over all space at a fixed time. Therefore, $H'_I(t)$ still depends on t . Time is subsequently integrated in the Dyson series:

$$\boxed{\int_{-\infty}^{\infty} dt H'_I(t) = \frac{\lambda}{4!} \int d^4 z \phi_I^4(z), \quad z := (t, \mathbf{z}).} \quad (43)$$

5 Starting point of the perturbative calculation

Having constructed ϕ_I and H'_I , our calculations begin with the Gell-Mann–Low formula⁴ for the two-point correlator. Indeed, this formula works precisely because it replaces correlators of the full field with correlators of the free Klein–Gordon field, together with the time-ordered insertion of the interaction.

$$\langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle = \frac{\langle 0 | T \{ \phi_I(x) \phi_I(y) e^{-i \int_{-\infty}^{\infty} dt H'_I(t)} \} | 0 \rangle}{\langle 0 | T \{ e^{-i \int_{-\infty}^{\infty} dt H'_I(t)} \} | 0 \rangle} := \frac{N}{D}. \quad (44)$$

Here $\phi(x)$ denotes $\phi_H(x)$, the full field in the Heisenberg picture; $|\Omega\rangle$ is the interacting vacuum and $|0\rangle$ the free vacuum, both normalized. The symbols N and D denote the numerator and denominator of (44), respectively. The formula is used perturbatively, with vacuum preparation and time limits defined by a convergence prescription compatible with Feynman conditions. The free vacuum is assumed to have nonzero overlap with the interacting vacuum in the regularized theory. The $i\epsilon$ fixes the pole prescription for propagators; it does not remove ultraviolet divergences, which require regularization and renormalization.

Substituting (40) into (44), the evolution operator contains, by (43), the exponential of $-i \frac{\lambda}{4!} \int d^4 z \phi_I^4(z)$ under time ordering. We first expand the exponential formally in its Taylor series:

$$\begin{aligned} \exp \left[-i \frac{\lambda}{4!} \int d^4 z \phi_I^4(z) \right] &= 1 - i \frac{\lambda}{4!} \int d^4 z \phi_I^4(z) \\ &\quad + \frac{1}{2!} \left(-i \frac{\lambda}{4!} \right)^2 \left[\int d^4 z \phi_I^4(z) \right]^2 + \dots \end{aligned} \quad (45)$$

In the quadratic term, each integral factor has its own integration variable. Since z is a dummy variable, we may call them z_1 and z_2 :

$$\begin{aligned} \left[\int d^4 z \phi_I^4(z) \right]^2 &= \left[\int d^4 z_1 \phi_I^4(z_1) \right] \left[\int d^4 z_2 \phi_I^4(z_2) \right] \\ &= \int d^4 z_1 \int d^4 z_2 \phi_I^4(z_1) \phi_I^4(z_2). \end{aligned} \quad (46)$$

The points z_1 and z_2 are integrated independently; squaring the integral produces fields evaluated at both points, not eight fields at a single point. At order n , there are n variables $z_1, \dots, z_n$, one for each integral factor. We now apply T to the fields in each term, ordering them according to their times $z_j = (t_j, \mathbf{z}_j)$, and obtain

$$\begin{aligned} T \left\{ \exp \left[-i \frac{\lambda}{4!} \int d^4 z \phi_I^4(z) \right] \right\} &= 1 - i \frac{\lambda}{4!} \int d^4 z_1 T \{ \phi_I^4(z_1) \} \\ &\quad + \frac{(-i\lambda)^2}{2!(4!)^2} \int d^4 z_1 \int d^4 z_2 T \{ \phi_I^4(z_1) \phi_I^4(z_2) \} + \dots \end{aligned} \quad (47)$$

Therefore,

$$N = \langle 0 | T \{ \phi_I(x) \phi_I(y) \} | 0 \rangle - i \frac{\lambda}{4!} \int d^4 z \langle 0 | T \{ \phi_I(x) \phi_I(y) \phi_I^4(z) \} | 0 \rangle + O(\lambda^2), \quad (48)$$

$$D = 1 - i \frac{\lambda}{4!} \int d^4 z \langle 0 | T \{ \phi_I^4(z) \} | 0 \rangle + O(\lambda^2). \quad (49)$$

The denominator D contains the vacuum bubbles. When forming the ratio N/D , these disconnected pieces cancel, ultimately leaving the contributions connected to the external pair x, y .

⁴Murray Gell-Mann and Francis E. Low introduced this result in the context of the perturbative formulation of quantum field theories. In these notes, we use it as the rule that allows expectation values in the interacting vacuum $|\Omega\rangle$ to be written as ratios of expectation values in the free vacuum $|0\rangle$.

6 Wick's theorem at first order

The basic contraction is the free propagator

$$\Delta_F(a-b) := \langle 0|T\{\phi_I(a)\phi_I(b)\}|0\rangle. \quad (50)$$

The contractions of the six-field term are grouped as

$$\boxed{\langle 0|T\{\phi_I(x)\phi_I(y)\phi_I^4(z)\}|0\rangle = 3\Delta_F(x-y)[\Delta_F(z-z)]^2 + 12\Delta_F(x-z)\Delta_F(y-z)\Delta_F(z-z).} \quad (51)$$

The factor 3 corresponds to contractions in which x is joined to y ; the factor 12 corresponds to contractions in which both external points are joined to the vertex.

For the denominator, Wick's theorem allows the four fields at z to be paired in three ways. If we temporarily label the fields 1, 2, 3, 4, the pairings are (12)(34), (13)(24), and (14)(23). They all give the same product of two propagators:

$$\langle 0|T\{\phi_I^4(z)\}|0\rangle = 3[\Delta_F(z-z)]^2, \quad \frac{3}{4!} = \frac{1}{8}. \quad (52)$$

From (51) and (52),

$$N = \Delta_F(x-y) - i\lambda \int d^4z \left[\frac{1}{8}\Delta_F(x-y)[\Delta_F(z-z)]^2 + \frac{1}{2}\Delta_F(x-z)\Delta_F(y-z)\Delta_F(z-z) \right] + O(\lambda^2), \quad (53)$$

$$D = 1 - i\frac{\lambda}{8} \int d^4z [\Delta_F(z-z)]^2 + O(\lambda^2). \quad (54)$$

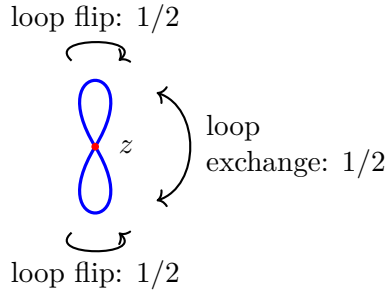


Figure 3. Figure-eight vacuum bubble: two loops joined at a vertex.

Its symmetry factor is $S = 2 \cdot 2 \cdot 2 = 8^5$: we can exchange the two ends of each loop and also exchange the loops with each other without changing the diagram. Each of these symmetries of order two contributes a factor 1/2:

$$\frac{1}{S} = \frac{1}{2} \frac{1}{2} \frac{1}{2} = \frac{1}{8}, \quad \mathcal{A}_{\text{vac}}^{(1)} = \frac{-i\lambda}{8} \int d^4z [\Delta_F(z-z)]^2. \quad (55)$$

Exchanging the ends of a loop is often visualized as a flip of the loop; what is counted is that discrete permutation, not an arbitrary rotation of the drawing. This is a single vacuum bubble with two loops, not two independent bubbles. The 1/8 obtained from Wick's theorem already includes these symmetries: one does not divide by 8 again.

⁵See K. Peeters and M. Zamaklar, *A guide to quantum field theory*, Section 4.3, the discussion of equation (4.34), pp. 31–32: the flips of each loop and the exchange of the two loops of the figure eight are counted. [Durham notes](#).

6.1 Cancellation of vacuum bubbles

Cancellation does not occur only at first order. Each diagram in the numerator separates into a part attached to the external points x, y and a collection of vacuum bubbles disconnected from them. These bubbles form exactly the same factor that appears in the denominator. Diagrammatically, the normalization is written as

$$\frac{N}{D} = \frac{\left[\text{---} + \text{---} \text{---} + \text{---} \text{---} + \dots \right] \exp \left(\text{---} + \dots \right)}{\exp \left(\text{---} + \dots \right)} \quad (56)$$

$$= \text{---} + \text{---} \text{---} + \text{---} \text{---} + \dots$$

Figure 4. Cancellation of vacuum bubbles in the normalized correlator.

Each drawing represents its full contribution, with the corresponding propagators, vertices, integrals, and symmetry factors. The endpoints of the open lines are x, y . The exponential contains the sum of connected vacuum bubbles; expanding it also generates collections of independent bubbles, with their factors $1/n!$.

After canceling the common vacuum-bubble factor in (56), the sum of diagrams connected to the external pair x, y remains. We assume a vacuum that preserves the symmetry $\phi \mapsto -\phi$, so one-field correlators vanish. Loops attached to the external points, such as the *tadpole* loop, remain in this sum.

To obtain the correlator through first order, we retain only the first two diagrams: free propagation, of order λ^0 , and the *tadpole*, of order λ . The chain and all other terms of second or higher order are included in $O(\lambda^2)$:

$$\frac{N}{D} = \text{---} + \text{---} \text{---} + O(\lambda^2). \quad (57)$$

Writing the contributions of these two diagrams in position space gives directly

$$\langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle = \Delta_F(x - y) - i \frac{\lambda}{2} \int d^4 z \Delta_F(x - z) \Delta_F(y - z) \Delta_F(z - z) + O(\lambda^2). \quad (58)$$

The zeroth-order term has $S = 1$, and the first-order *tadpole* has $S = 2$. The three second-order topologies are developed in the next section. Here $\Delta_F(z - z) = \Delta_F(0)$ is formally ultraviolet divergent and must be regularized and renormalized.⁶ The symmetry factors are unchanged.

⁶With our convention,

$$\Delta_F(0) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2\sqrt{\mathbf{p}^2 + m^2}}. \quad (IV)$$

At large momenta, the radial measure turns the integral into an expression that grows as $\int p dp$: hence the ultraviolet divergence, associated with short wavelengths. A cutoff $|\mathbf{p}| \leq \Lambda$ regularizes the integral:

$$\Delta_F^{(\Lambda)}(0) = \frac{1}{4\pi^2} \int_0^\Lambda \frac{p^2 dp}{\sqrt{p^2 + m^2}} \sim \frac{\Lambda^2}{8\pi^2}, \quad \Lambda \gg m. \quad (V)$$

Renormalization adjusts parameters and counterterms through physical conditions to obtain finite predictions when the regulator is removed. In this tadpole, the divergence is absorbed into the mass counterterm; it is not a vacuum bubble that cancels, nor does it change $S = 2$. See Peeters and Zamaklar, [A guide to quantum field theory](#), Sections 7.1 and 7.2.

7 Connected contributions at second order

For the interaction $\lambda\phi^4/4!$ without normal ordering, the three connected topologies of the two-point correlator at second order are as follows:

Two *tadpole* diagrams (“tadpoles”) in a chain:

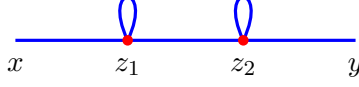


Figure 5. Two *tadpole* diagrams in a chain.

$$\frac{(-i\lambda)^2}{4} \int d^4 z_1 \int d^4 z_2 \Delta_F(x - z_1) \Delta_F(z_1 - z_1) \times \Delta_F(z_1 - z_2) \Delta_F(z_2 - z_2) \Delta_F(z_2 - y). \quad (59)$$

Two lines between vertices and a local loop:

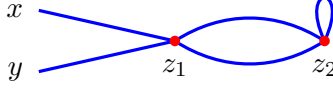


Figure 6. Two lines between vertices and a local loop.

$$\frac{(-i\lambda)^2}{4} \int d^4 z_1 \int d^4 z_2 \Delta_F(x - z_1) \Delta_F(y - z_1) [\Delta_F(z_1 - z_2)]^2 \Delta_F(z_2 - z_2). \quad (60)$$

Sunset diagram:

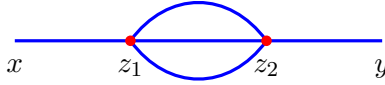


Figure 7. *Sunset* diagram.

$$\frac{(-i\lambda)^2}{6} \int d^4 z_1 \int d^4 z_2 \Delta_F(x - z_1) [\Delta_F(z_1 - z_2)]^3 \Delta_F(z_2 - y). \quad (61)$$

The denominators 4, 4, and 6 are the symmetry factors of these topologies, respectively. They can also be obtained by counting the contractions with the external points x, y fixed: there are 288, 288, and 192 pairings, respectively. Dividing by the factor $2!(4!)^2 = 1152$ from the Dyson expansion gives $1/4$, $1/4$, and $1/6$.

8 Feynman rules in position space

The preceding derivations give three rules for the interaction $\lambda\phi^4/4!$:

1. **Lines.** Each contraction between points a and b contributes the propagator $\Delta_F(a - b)$, as in .

- 2. Vertices.** Each vertex has four line ends and contributes $(-i\lambda) \int d^4z$. Its position z is integrated, while the external points x, y stay fixed. With V vertices there are V independent integrations.
- 3. Symmetries.** Divide by the factor S , counting the permutations that preserve the diagram and fix the external points. For example, the *tadpole*  has $S = 2$; the three second-order topologies have $S = 4, 4, 6$.

For a diagram with V vertices, the three rules are summarized by

$$\boxed{\frac{1}{S}(-i\lambda)^V \int \prod_{j=1}^V d^4z_j \prod_{\text{lines } (ab)} \Delta_F(a-b).} \quad (62)$$

To obtain the normalized correlator, sum the diagrams connected to the external pair x, y through the desired order, keeping those points fixed. Vacuum bubbles have already canceled according to (56); loops attached to the external points remain. The factor $1/S$ accounts for diagram symmetries and must not be applied again when the coefficient has already been obtained by counting Wick contractions.